

JONATHAN NADEAU

DATA ANALYST

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Portfolio Website: nadeaujonny.github.io

SUMMARY

Mathematics graduate (UCSB) and Google Data Analytics certified analyst, sole developer of six internal data tools at a defense and telecom electronics manufacturer. Work spans reverse-engineering undocumented legacy systems and reporting processes, auditing historical records against their own source data, live vendor API integration, and test-validated Python applications shipped to engineering and procurement users, plus 1,500+ hours translating quantitative material for non-technical audiences.

TECHNICAL SKILLS

SQL & Databases: Oracle SQL, BigQuery, MySQL, SQLite · CTEs, window functions, stored procedures, DDL/DML, relational schema mapping

Python & R: pandas, NumPy, scikit-learn, openpyxl, Streamlit, pytest, Tkinter, PyInstaller, matplotlib · R (tidyverse)

Spreadsheets & BI: Excel (Power Query, Power Pivot, DAX, dynamic arrays, Solver), Power BI, Tableau, Looker Studio

Data & Process: ETL pipelines, REST API and OAuth2 integration, data validation and reconciliation, data-quality profiling, regression and golden-file testing, requirements gathering, technical documentation, Git/GitHub

PROFESSIONAL EXPERIENCE

Data Analyst Intern | Escape Communications

July 2026 – Present | Torrance, CA

Sole developer on six internal tools across engineering, quality, and procurement at a defense and telecom hardware manufacturer; owned requirements, architecture, validation, documentation, and deployment on each.

- **Failure Analysis Search Tool:** Extracted 4,038 hardware failure records (2007–2026) and 13,252 engineer comments from an undocumented legacy Bugzilla instance via REST API, then indexed 2,245 distinct PCB reference designators out of free text with a standard-library-only Python pipeline, validating 11 component prefixes by manual sampling and dropping one after measuring an ~85% false-positive rate. Shipped a Streamlit search app with TF-IDF similarity and watermark-tracked incremental refresh, plus runbook, user guide, and deployment docs.
- **BOM Checker (published to the company share):** Built and packaged a standalone Windows BOM-auditing tool (Python, pandas, openpyxl, PyInstaller; 427 automated tests) flagging 16 classes of structural and parametric defect in raw PADS Logic exports while returning the user's original worksheet byte-for-byte unchanged, verified cell-by-cell. Wrote an offline decoder for manufacturers' part-numbering schemes to check BOM-stated specs against the part actually specified; validated across 8 production BOMs (5 product lines, 2,290 rows, 869 part numbers), where 9 of 10 flagged mismatches were confirmed as genuine errors against manufacturer datasheets.
- **BOM Sourcing & Purchase Planning Tool:** Built a 703-test Python pipeline, frozen to a Windows executable, that prices and risk-assesses 300+ line BOMs against live DigiKey (OAuth2) and Mouser APIs behind a vendor-agnostic layer with TTL caching, rate limiting, and daily quota accounting, outputting an Excel workbook covering obsolescence risk, build cost, and restock timing. Diagnosed a suspected API data-reliability problem as a packaging-variant selection defect by measuring 266 rows against both vendors' own tools; the fix raised priced coverage from 65 to 195 of 266 rows.
- **FPGA Waveform Capture Validation Tool:** Replaced by-eye waveform inspection in a hardware verification workflow with an automated validator (Python, Tkinter, PyInstaller) that reverse-engineers the vendor capture archive, reconstructs the message stream from hardware handshake signals, and reports 13 named checks with the exact word and sample index of any mismatch. Built an 11-scenario synthetic capture generator to prove the failure paths fire, and surfaced two undocumented hardware handshake behaviors by investigating a check that contradicted known-good data.
- **Inventory Audit Log Tool:** Turned a one-page requirements memo from the VP of Engineering into a shipped Windows application merging dated physical count sheets into one running audit log, resolving underspecified behaviors (blank versus zero counts, duplicate parts, same-date collisions) as explicit stakeholder decisions before implementation. Append-only semantics and atomic in-place writes guarantee a failed run leaves the record byte-identical, verified by 69 tests; delivered from written spec in under two days.
- **ERP Quality Reporting Audit (PEDYN):** Reverse-engineered six undocumented ISO 9001 management-review reports from 20 archived workbooks and 16 ERP table schemas after the original author's departure, proving the reconstruction by reproducing four consecutive years of output exactly (281/146/106/90 records). Surfaced four material errors in the audit record, including a figure recorded as 338 against an actual 101 and carried forward two years, plus seven undocumented metric-definition changes; authored the operating spec for an automated replacement projected to remove 85–95% of manual effort on five of six reports.

Mathematics Instructor & Tutor | Mr. J's Academy · Wyzant · Blue Train College Prep

Sept 2022 – May 2025 | Torrance & Goleta, CA

- Delivered 1,500+ hours of instruction spanning Algebra 1 through Calculus and SAT prep across 17+ recurring clients, maintaining a 4.8-star Wyzant rating and 100% client retention; promoted to high school lead tutor at Mr. J's Academy with a 50% compensation increase. Diagnosed learning gaps from assessment data and built targeted problem sets per student.

Lab Technician | Escape Communications

Summers 2017 – 2023, 2025 | Torrance, CA

- Performed QA testing and calibration on power supply units, recording multimeter measurements into Excel trackers built to flag out-of-tolerance values automatically; tracked component inventory across product lines and tested completed boards for functional compliance.

SELECTED PORTFOLIO PROJECTS (15 deployed at nadeaujonny.github.io)

- **Supply Chain Analytics (Oracle SQL, Excel, Power BI):** 180K orders; stored procedures with scheduled automation, Power Query ETL, ABC/XYZ classification, ETS forecasting, MRP simulation, and a 6-page star-schema Power BI dashboard diagnosing a structural 42.7% on-time delivery rate.
- **Skechers FP&A Model (Excel):** Three-statement model built from FY2020–FY2024 10-K filings with a segment-level revenue engine, Bear/Base/Bull scenarios, sensitivity tables, and a KPI dashboard; all balance-sheet and cash-flow audit checks passing in every forecast year.
- **Customer Churn Prediction (Python, scikit-learn, Streamlit):** Compared five classifiers on 7K records with recall selected as the priority metric on a stated cost asymmetry, tuned via RandomizedSearchCV, explained predictions with SHAP, and deployed a scoring app at 0.83 AUC and 79% recall.

EDUCATION & CERTIFICATIONS

B.A. in Mathematics | University of California, Santa Barbara

Sept 2018 – March 2023

GPA 3.40 (Mathematics GPA 3.49) · 4x Dean's Honors · Minor: Secondary Mathematics & Science Education

Data Science Certificate | UCLA Extension

June 2026 – Dec 2026 (in progress)

Intro to Data Science · Exploratory Data Analysis & Visualization · Big Data Management · Machine Learning Using Python

Google Data Analytics Professional Certificate | Coursera

Oct 2025 – Jan 2026 · 100% average, all nine courses

Single Subject Mathematics Teaching Credential Coursework | CSU Long Beach

Aug 2024 – Oct 2025 · GPA 4.00